

Annotations Using ggplot2 - Hotel Booking Dataset

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Background

In this notebook, we'll review a scenario, and add annotations to a data visualization with ggplot2. We will also save images from ggplot2 visualizations so that we can add them directly to presentations.

The Scenario

As junior data analysts for a hotel booking company, we have been creating visualizations in R with the ggplot2 package to share insights about our data with stakeholders. After creating a series of visualizations using ggplot(), ggplot2 aesthetics, and filters, our stakeholder asks we to add annotations to our visualizations to help explain our findings in a presentation. Luckily, ggplot2 has annotation functions built in.

Step 1: Import our data

The data in this example is originally from the article Hotel Booking Demand Datasets

(<https://www.sciencedirect.com/science/article/pii/S2352340918315191>

(<https://www.sciencedirect.com/science/article/pii/S2352340918315191>)), written by Nuno Antonio, Ana Almeida, and Luis Nunes for Data in Brief, Volume 22, February 2019.

The data was downloaded and cleaned by Thomas Mock and Antoine Bichat for #TidyTuesday during the week of February 11th, 2020 (<https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md> (<https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md>)).

We can learn more about the dataset here: <https://www.kaggle.com/jessemostipak/hotel-booking-demand> (<https://www.kaggle.com/jessemostipak/hotel-booking-demand>)

Feed in the file 'hotel_bookings.csv' into a data frame:

```
hotel_bookings <- read.csv("hotel_bookings.csv")
```

Step 2: Refresh Our Memory

By now, we are pretty familiar with this data set. But we can refresh our memory with the head() and colnames() functions.

```
head(hotel_bookings)
```

```

##      hotel is_canceled lead_time arrival_date_year arrival_date_month
## 1 Resort Hotel      0      342      2015      July
## 2 Resort Hotel      0      737      2015      July
## 3 Resort Hotel      0       7      2015      July
## 4 Resort Hotel      0      13      2015      July
## 5 Resort Hotel      0      14      2015      July
## 6 Resort Hotel      0      14      2015      July
## arrival_date_week_number arrival_date_day_of_month stays_in_weekend_nights
## 1      27      1      0
## 2      27      1      0
## 3      27      1      0
## 4      27      1      0
## 5      27      1      0
## 6      27      1      0
## stays_in_week_nights adults children babies meal country market_segment
## 1      0      2      0      0 BB PRT Direct
## 2      0      2      0      0 BB PRT Direct
## 3      1      1      0      0 BB GBR Direct
## 4      1      1      0      0 BB GBR Corporate
## 5      2      2      0      0 BB GBR Online TA
## 6      2      2      0      0 BB GBR Online TA
## distribution_channel is_repeated_guest previous_cancellations
## 1      Direct      0      0
## 2      Direct      0      0
## 3      Direct      0      0
## 4      Corporate      0      0
## 5      TA/T0      0      0
## 6      TA/T0      0      0
## previous_bookings_not_canceled reserved_room_type assigned_room_type
## 1      0      C      C
## 2      0      C      C
## 3      0      A      C
## 4      0      A      A
## 5      0      A      A
## 6      0      A      A
## booking_changes deposit_type agent company days_in_waiting_list customer_type
## 1      3 No Deposit NULL NULL 0 Transient
## 2      4 No Deposit NULL NULL 0 Transient
## 3      0 No Deposit NULL NULL 0 Transient
## 4      0 No Deposit 304 NULL 0 Transient
## 5      0 No Deposit 240 NULL 0 Transient
## 6      0 No Deposit 240 NULL 0 Transient
## adr required_car_parking_spaces total_of_special_requests reservation_status
## 1 0      0      0 Check-Out
## 2 0      0      0 Check-Out
## 3 75     0      0 Check-Out
## 4 75     0      0 Check-Out
## 5 98     0      1 Check-Out
## 6 98     0      1 Check-Out
## reservation_status_date
## 1      2015-07-01
## 2      2015-07-01

```

```
## 3      2015-07-02
## 4      2015-07-02
## 5      2015-07-03
## 6      2015-07-03
```

```
colnames(hotel_bookings)
```

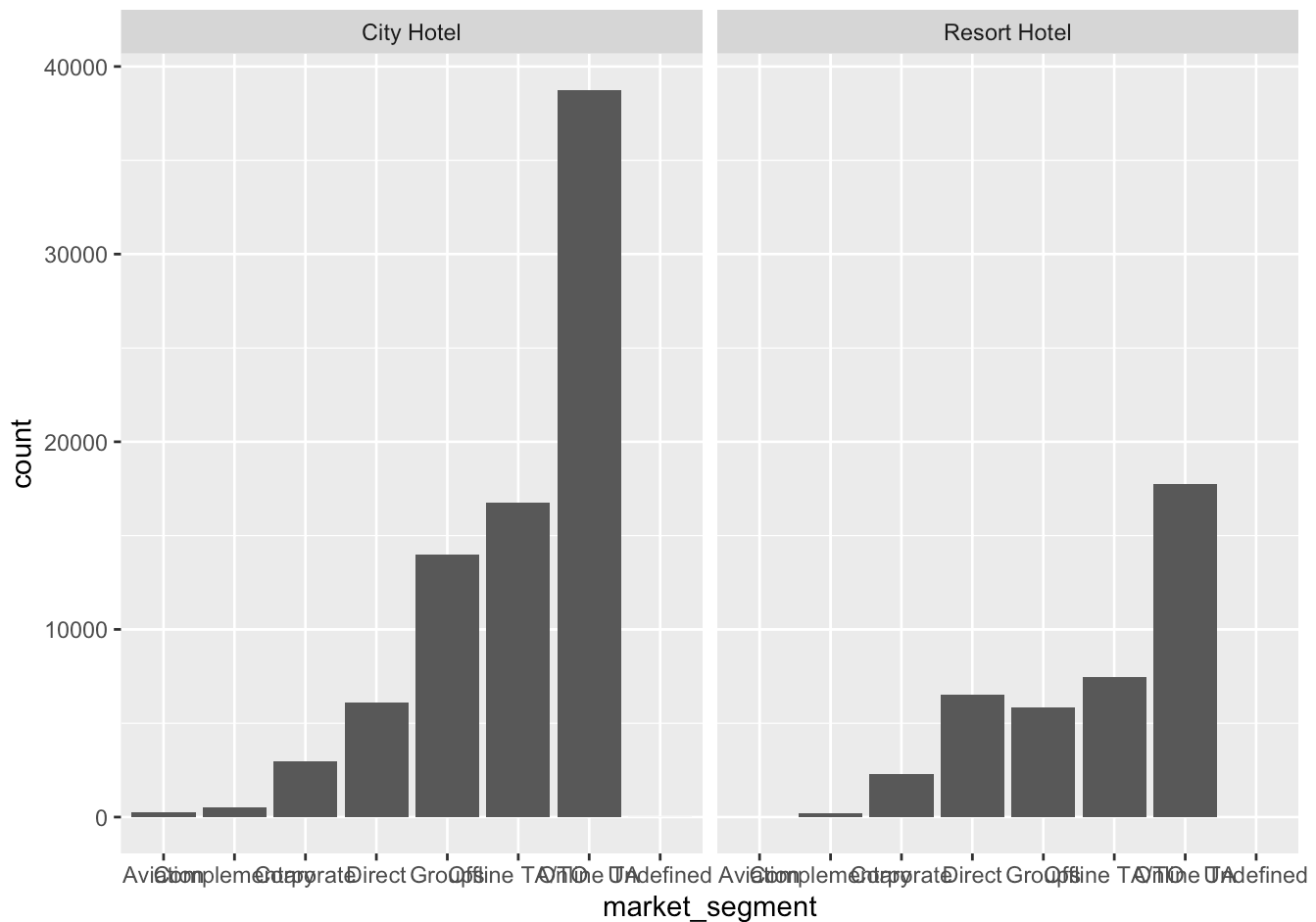
```
## [1] "hotel" "is_canceled"
## [3] "lead_time" "arrival_date_year"
## [5] "arrival_date_month" "arrival_date_week_number"
## [7] "arrival_date_day_of_month" "stays_in_weekend_nights"
## [9] "stays_in_week_nights" "adults"
## [11] "children" "babies"
## [13] "meal" "country"
## [15] "market_segment" "distribution_channel"
## [17] "is_repeated_guest" "previous_cancellations"
## [19] "previous_bookings_not_canceled" "reserved_room_type"
## [21] "assigned_room_type" "booking_changes"
## [23] "deposit_type" "agent"
## [25] "company" "days_in_waiting_list"
## [27] "customer_type" "adr"
## [29] "required_car_parking_spaces" "total_of_special_requests"
## [31] "reservation_status" "reservation_status_date"
```

Step 3: Install and load the ‘ggplot2’ package

Step 4: Annotating our chart

As a refresher, here is the chart we created before:

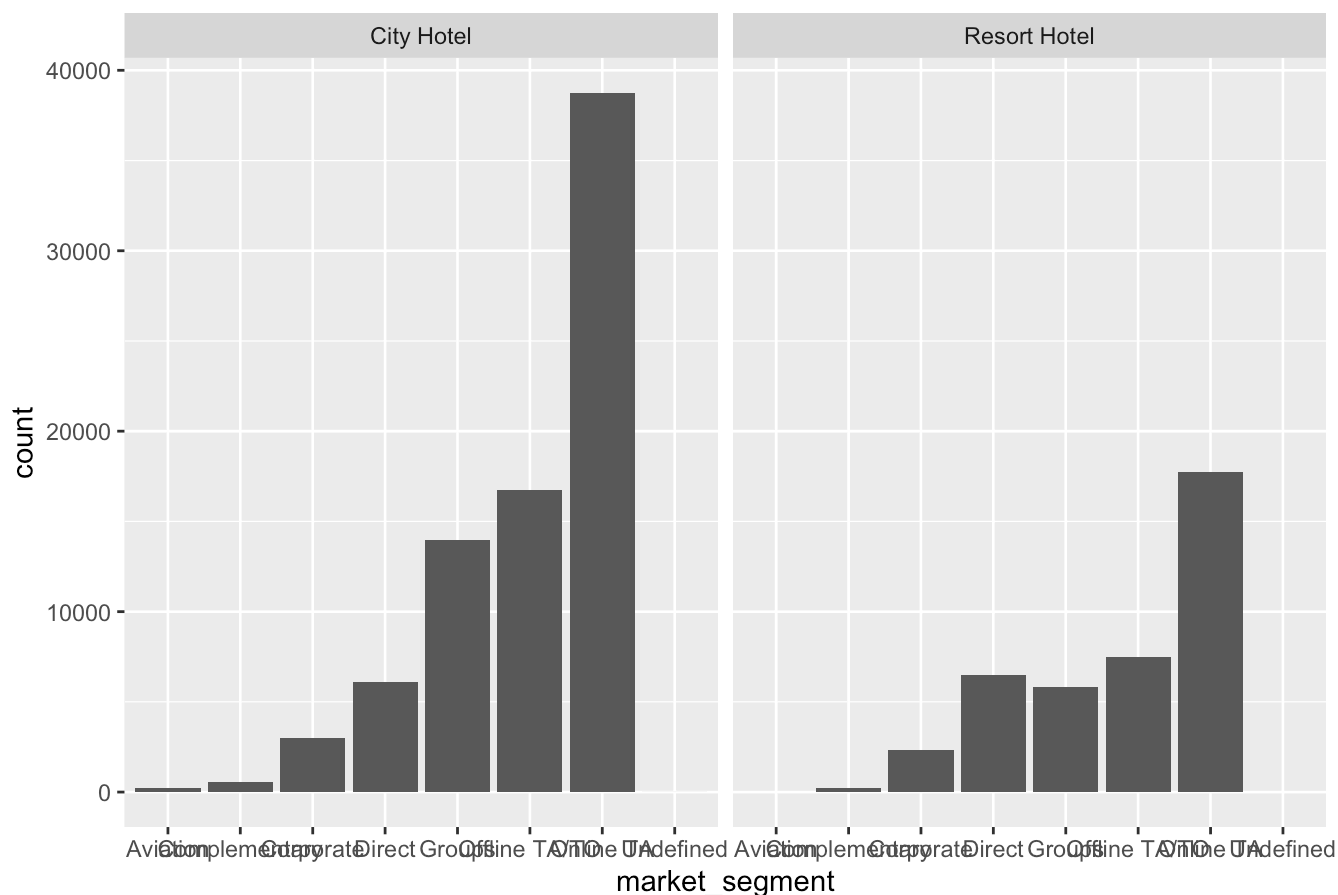
```
ggplot(data = hotel_bookings) +
  geom_bar(mapping = aes(x = market_segment)) +
  facet_wrap(~hotel)
```



The first step will be adding a title; that is often the first thing people will pay attention to when they encounter a data visualization for the first time. To add a title, we will add `labs()` at the end of our `ggplot()` command and then input a title there:

```
ggplot(data = hotel_bookings) +
  geom_bar(mapping = aes(x = market_segment)) +
  facet_wrap(~hotel) +
  labs(title="Comparison of market segments by hotel type for hotel bookings")
```

Comparison of market segments by hotel type for hotel bookings



We also want to add another detail about what time period this data covers. To do this, we need to find out when the data is from.

We realize we can use the `min()` function on the year column in the data:

```
min(hotel_bookings$arrival_date_year)
```

```
## [1] 2015
```

And the `max()` function:

```
max(hotel_bookings$arrival_date_year)
```

```
## [1] 2017
```

But we will need to save them as variables in order to easily use them in our labeling:

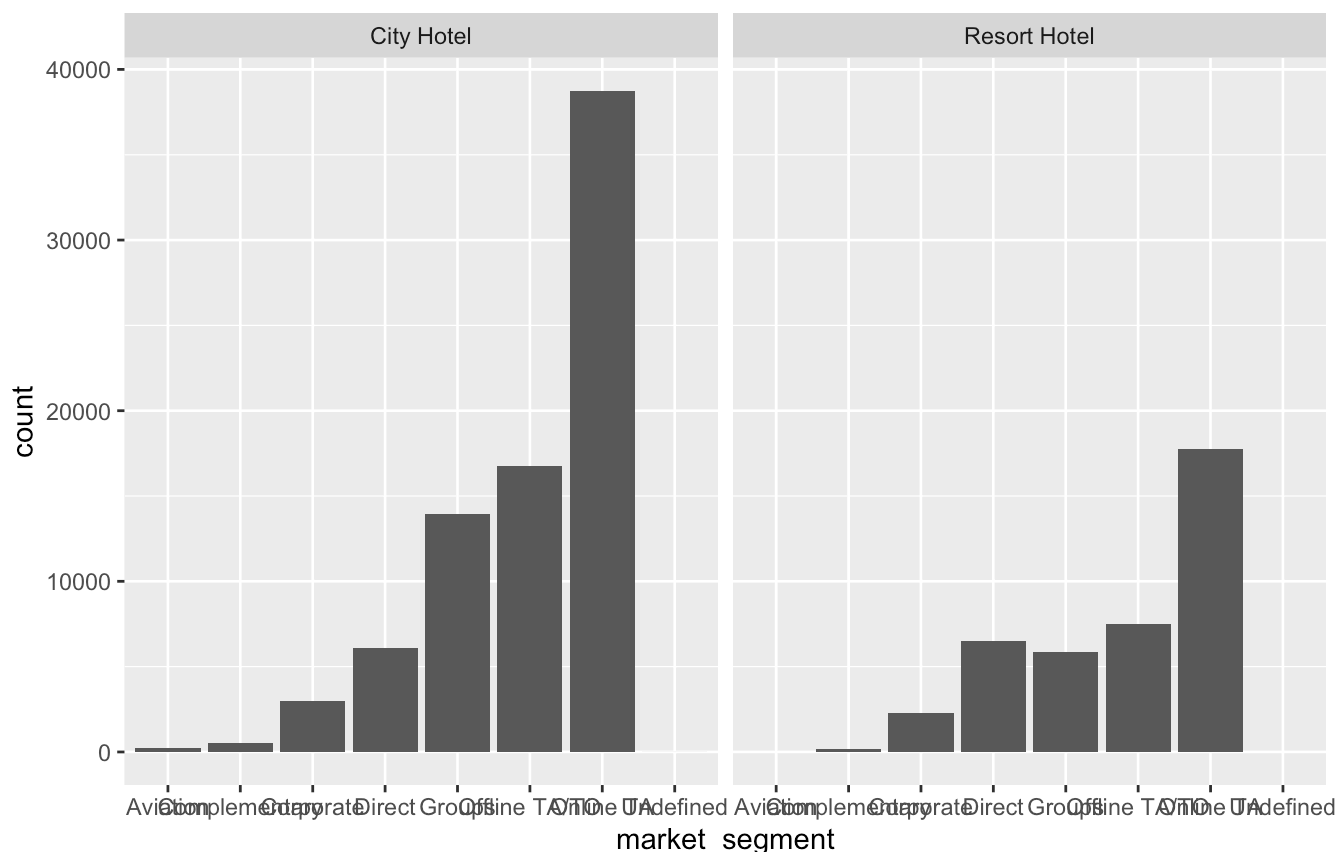
```
mindate <- min(hotel_bookings$arrival_date_year)
maxdate <- max(hotel_bookings$arrival_date_year)
```

Now, we will add in a subtitle using `subtitle=` in the `labs()` function. Then, we can use the `paste0()` function to use our newly-created variables in our labels. This is really handy, because if the data gets updated and there is more recent data added, we don't have to change the code below because the variables are dynamic:

```
ggplot(data = hotel_bookings) +
  geom_bar(mapping = aes(x = market_segment)) +
  facet_wrap(~hotel) +
  labs(title="Comparison of market segments by hotel type for hotel bookings",
       subtitle=paste0("Data from: ", mindate, " to ", maxdate))
```

Comparison of market segments by hotel type for hotel bookings

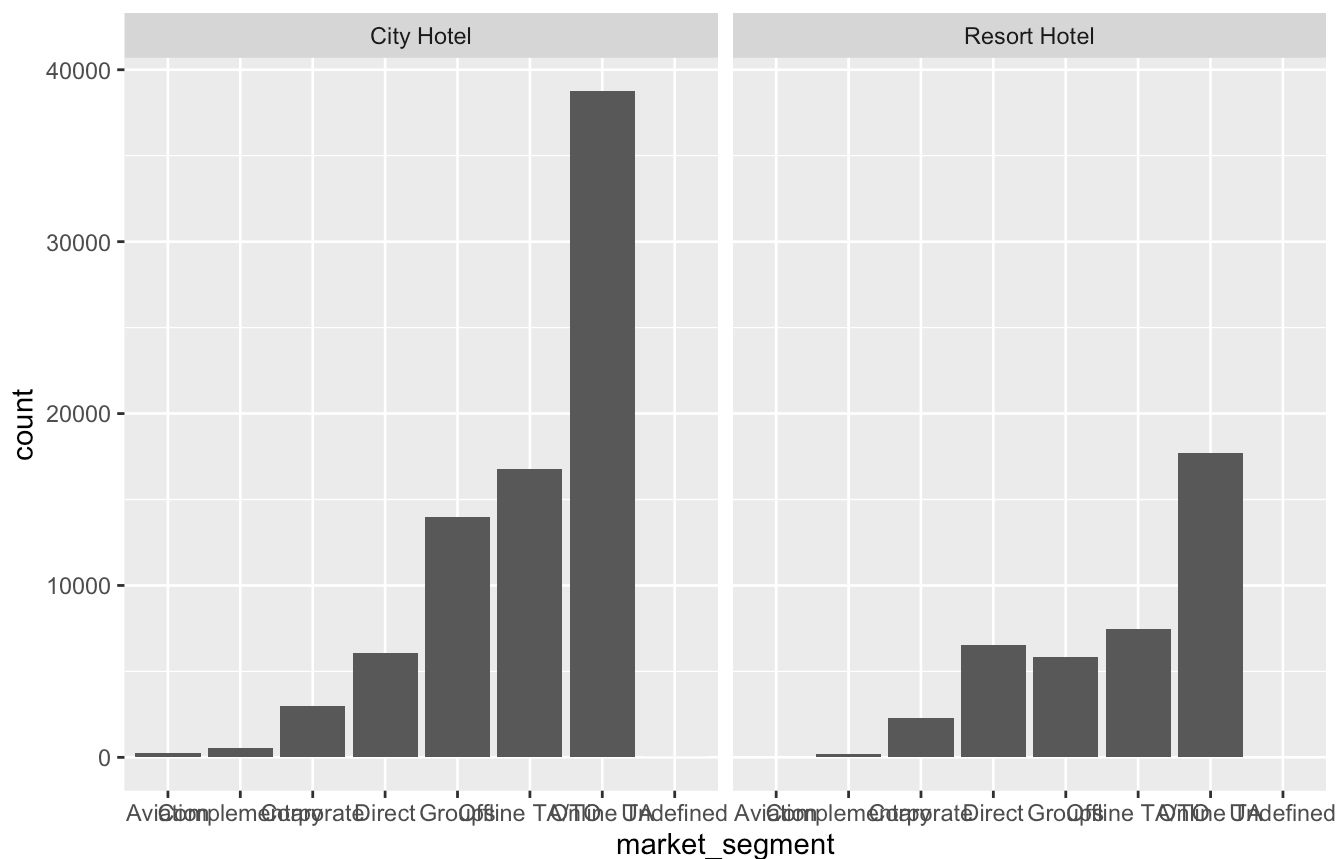
Data from: 2015 to 2017



We decide to switch the subtitle to a caption which will appear in the bottom right corner instead.

```
ggplot(data = hotel_bookings) +
  geom_bar(mapping = aes(x = market_segment)) +
  facet_wrap(~hotel) +
  labs(title="Comparison of market segments by hotel type for hotel bookings",
       caption=paste0("Data from: ", mindate, " to ", maxdate))
```

Comparison of market segments by hotel type for hotel bookings

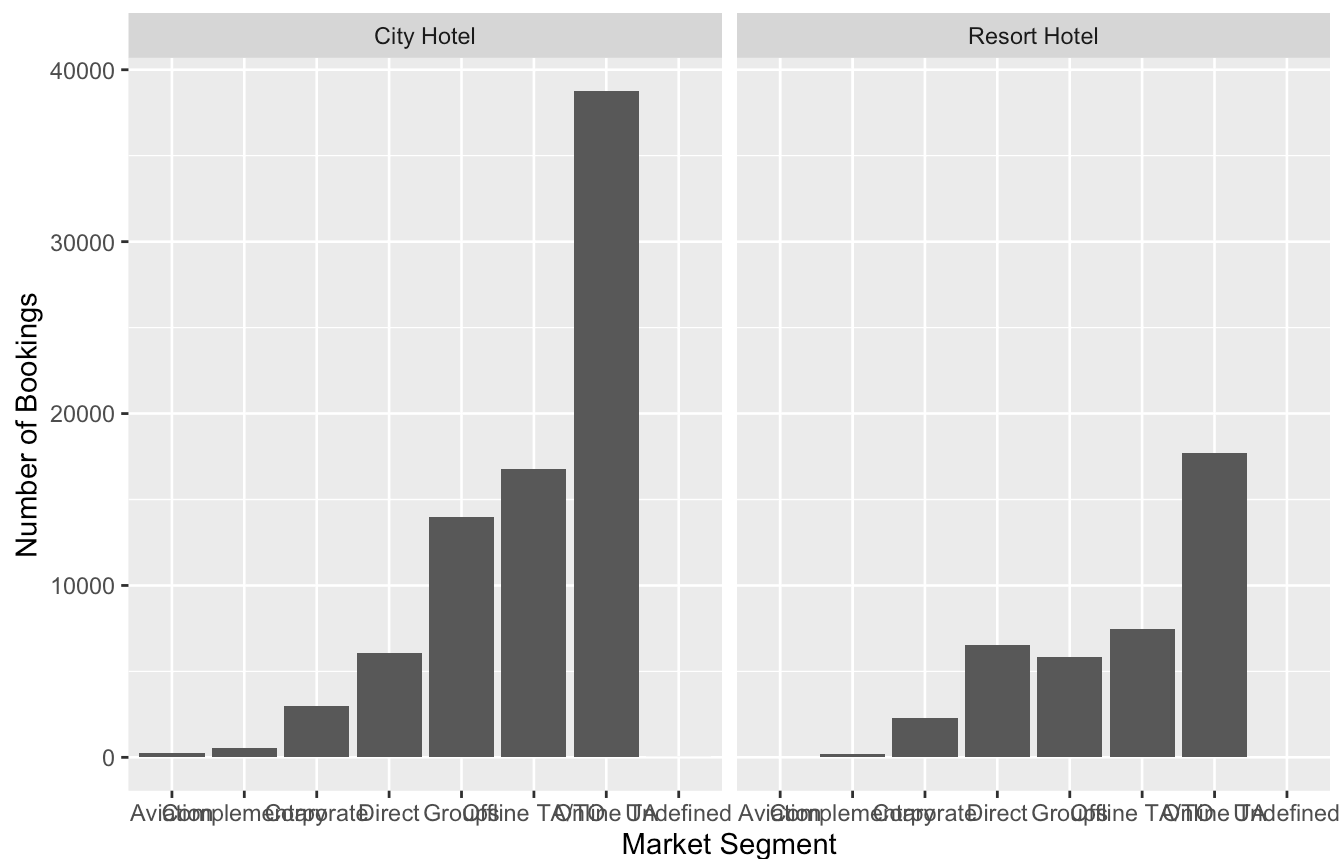


Data from: 2015 to 2017

Now we want to clean up the x and y axis labels to make sure they are really clear. To do that, we can add to the `labs()` function and use `x=` and `y=`.

```
ggplot(data = hotel_bookings) +
  geom_bar(mapping = aes(x = market_segment)) +
  facet_wrap(~hotel) +
  labs(title="Comparison of market segments by hotel type for hotel bookings",
       caption=paste0("Data from: ", mindate, " to ", maxdate),
       x="Market Segment",
       y="Number of Bookings")
```

Comparison of market segments by hotel type for hotel bookings



Data from: 2015 to 2017

Step 5: Saving our chart

The `ggsave()` function was used to save the last plot that was generated, so if we have run something after running the code chunk above, then run that code chunk again.

Then we save that plot as a .png file named `city_payment_chart`, which makes it clear to our stakeholders what the .png file contains.

```
ggsave('hotel_booking_chart.png')
```

```
## Saving 7 x 5 in image
```

Tip

The default dimensions that `ggsave()` saved our image as are 7x7.

We can see these dimensions listed after we run the code chunk.

If we wanted to make our chart bigger and more rectangular to fit the slide show presentation, we could specify the height and width of our .png in the `ggsave()` command. Let's create a 16x8 .png image:

```
ggsave('hotel_booking_chart.png',
      width=16,
      height=8)
```


Wrap Up

Now we have finished creating and exporting a data visualization with annotations in `ggplot2`, we can share what we created with key stakeholders to give them insights into our data findings.